

EN 510: Electrochemical Systems for Energy Engineering

Course Code	EN 510
Course Name	Electrochemical Systems for Energy Engineering
L-T-P-C	3-0-2-4
Prerequisites	None
Intended for	UG/PG (Compulsory for MTech. in Materials and Energy Engineering, and Elective for others)
Mutual Exclusion	None .
Approval	44th BoA (Nov 2021)

Course Contents

- **Electrochemical Engineering Fundamentals:** Electrical Current/Voltage, Faraday's Laws; Electric Efficiency, and Mass Balance; Electrode Potentials and Electrode–Electrolyte Interfaces; Potential Difference; Electrochemical Cells- Galvanic, Electrolytic and concentration. [5 Hours]
- **Thermodynamics and Kinetics of an Electrochemical Cell:** Electrochemical Cell Phases; The Nernst Equation; Mass Transfer Modes; Electrode Kinetics (Charger Transfer (Butler–Volmer Equation) and Mass Transfer (Diffusion Laws)); Limitations of Butler–Volmer Equation; Limiting Current Density; Galvanostatic Polarization; Polarization Methods- Linear Polarization, Tafel Extrapolation. [9 Hours]
- **Batteries:** Introduction; Basic Li battery; Lead acid battery; Nickel-Metal Hydride (Ni-MH) Rechargeable Batteries; Metal–Air batteries; Self-discharge of batteries; Jump starting a car; Battery safety and toxicity. [6 Hours]
- **Fuel Cells:** Introduction; Variety of fuel cells- proton exchange membrane fuel cell, Solid oxide fuel cell, Direct methanol fuel cell, Alkaline fuel cells; Hybrid fuel cell- battery system; Hydrogen Storage [6 Hours]
- **Supercapacitor:** Introduction; Electric double-layer capacitors (EDLCs); Pseudocapacitor; Asymmetric hybrid capacitors; Concerns with cell assembly; Energy density and power density. [5 Hours]
- **Electroanalytical methods:** Cyclic voltammetry and linear sweep voltammetry; The need for a reference electrode; Impedance Spectroscopy; Chronoamperometry; The open circuit potential; Galvanometric charge-discharge; Ring disc electrode [8 Hours]
- **Electrochemical manufacturing:** Electroplating; Electroless plating; Electrochemical machining and polishing. [3 Hours]

Laboratory/practical/tutorial Modules: [28 Hours]

- Experiment 1: Sample preparation
- Experiment 2: Cyclic voltammetry and linear sweep voltammetry
- Experiment 3: Impedance Spectroscopy
- Experiment 4: Chronoamperometry
- Experiment 5: Galvanometric charge-discharge
- Experiment 6: Ring disc electrode

Text Books

1. Zhang, J., Zhang, L., Liu, H., Sun, A., & Liu, R. S. (Eds.), **Electrochemical Technologies for Energy Storage and Conversion**, 2 Volume Set (Vol. 1). John Wiley & Sons, 2011.
2. Braun, A., **Electrochemical Energy Systems**, de Gruyter, 2018

References

Dicks, A. L., & Rand, D. A. (2018). Fuel cell systems explained. John Wiley & Sons.